The Lazaros–Eudora Method (LEM): A Topological Framework for User–LLM Interaction Dynamics

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Abstract

We introduce the Lazaros–Eudora Method (LEM), a conceptual and computational framework for analyzing user–LLM interactions as dynamic trajectories in latent representation space rather than as isolated prompt–response pairs. The central hypothesis of LEM is that repeated interaction between a user and a language model induces structured trajectory regimes that can be described through user-specific attractors and compact interaction signatures. We argue that purely geometric similarity is insufficient for characterizing such regimes and motivate a topological reformulation based on persistent homology. To render the framework testable without reliance on proprietary frontier-model internals, we present two toy-model validations in controlled synthetic settings. The first demonstrates geometric identifiability of user-conditioned trajectories in a controlled latent dynamical system, achieving perfect nearest-neighbor re-identification in the reported synthetic setting across N=500 users and ten independent random seeds, compared to a random baseline of 0.002. The second shows that convergent and cyclic interaction regimes can be distinguished through persistent H_1 structure after noisy projection into a higher-dimensional latent space, yielding a signal-to-noise ratio of 161.6. We position LEM as a testable research program for studying user–LLM interaction dynamics, while explicitly delimiting its current scope and empirical constraints.

1 Introduction

Current large language model evaluation remains dominated by static, turn-local metrics. These metrics can assess answer quality, benchmark performance, or stylistic alignment, but they do not directly describe the temporal structure of repeated user–model interaction. As a result, an important object remains under-modeled: the trajectory induced by a specific user interacting with a model across multiple turns.

The Lazaros–Eudora Method (LEM) is a framework designed to study this object. Its central premise is that user–LLM interaction should be modeled as a dynamical trajectory in latent space rather than as a collection of isolated prompt embeddings or output tokens. In this view, repeated interaction may generate structured regimes—for example convergence, recurrence, or instability—that are more naturally described in the language of dynamical systems and topology than in the language of static vector similarity alone.

This paper makes a limited but concrete contribution. It does not claim to establish a full theory of human cognition or a validated description of real frontier-model internals. Instead, it contributes:

1. a compact conceptual formalization of LEM;
2. a distinction between geometric and topological notions of interaction signatures;
3. two toy-model validations showing how user-conditioned attractors and trajectory classes can be operationalized in controlled synthetic settings.

The intended contribution is therefore methodological: to provide a testable framework for studying user–LLM interaction dynamics and to identify which parts of the theory are already operational, which remain open, and which are presently speculative.

2 Conceptual Framework

2.1 Interaction trajectories

We define an interaction trajectory as a time-ordered sequence of states

$$x_1, x_2, \ldots, x_T \in \mathcal{X},$$

where $\mathcal{X}$ is a latent or constructed representation space associated with a multi-turn user–LLM interaction. In practice, $\mathcal{X}$ may refer to internal activations, reduced coordinates, or synthetic latent states in a toy model.

This shift from isolated states to trajectories is the first core move of LEM. The object of study is not merely the embedding of a prompt or response, but the path induced by repeated interaction.

2.2 Cognitive signatures

A *cognitive signature* is a compact representation derived from an interaction trajectory that preserves salient structural properties of user–model dynamics. In the present paper, we distinguish two operational levels:

1. **Geometric signatures**, defined through distances, directional similarity, and nearest-neighbor re-identification in a controlled latent system;
2. **Topological signatures**, defined through persistent homology, especially the presence or absence of robust H_1 features.

This distinction is essential. Geometric signatures capture identifiability in a vector-space sense. Topological signatures capture regime structure that may not be reducible to distance alone.

2.3 Attractor-like interaction regimes

LEM uses the term *attractor* in a modeling sense. The claim is not that real human cognition has been recovered from LLM traces, but that repeated interaction can be approximated as a dynamical process whose trajectories may approach stable regimes. In the minimal setting considered here, these include:

- **Convergent regimes**, approximated by a stable fixed point;
- **Cyclic regimes**, approximated by a stable limit cycle;
- **Unstable or noisy regimes**, in which no compact structural summary is stable.

2.4 Core hypotheses

We focus on four hypotheses, of which only the first three are partially operationalized in the current paper.

H1: Geometric identifiability. User-conditioned interaction dynamics can generate trajectories that are re-identifiable above chance in a controlled latent system.

H2: Regime differentiation. Convergent and cyclic interaction regimes generate structurally distinct trajectory classes.

H3: Topological utility. Persistent homology captures structural differences between such regimes, especially through the presence or absence of persistent H_1 features.

H4: Cross-version stability. User-specific interaction signatures may remain partially stable across related model variants or model families. In this paper, H4 is treated as a theoretical target rather than an empirically established claim.

3 From Geometric Similarity to Topological Structure

Many current approaches to personalization, user modeling, and dialogue adaptation rely on static similarity measures, clustering, or compressed user embeddings. These approaches are useful, but they typically summarize multi-turn interaction as a point estimate or a low-order statistic.

LEM instead treats interaction as a trajectory and therefore distinguishes between two levels of structure:

1. **Geometric structure**: whether trajectories lie near each other or near user-conditioned attractors;
2. **Topological structure**: whether trajectories exhibit recurrence, loop-like organization, or topological triviality.

This motivates the use of topological data analysis (TDA), especially persistent homology. In the current work, the key topological observable is persistent H_1 structure, interpreted as evidence of recurrent cyclic organization in trajectory point clouds.

We explicitly do *not* claim that topological features directly reveal psychological essence or consciousness. The intended claim is narrower: in a controlled setting, topology can distinguish classes of interaction dynamics that are poorly summarized by static geometry alone.

4 Toy-Model Numerical Validation

The present paper does not attempt to validate LEM directly on proprietary frontier-model internals. Instead, it follows a more modest and methodologically transparent strategy: we test whether the core conceptual claims of the framework can be operationalized in controlled synthetic systems. The goal of these toy experiments is not realism, but falsifiability. If the central vocabulary of LEM—user-conditioned trajectories, attractor-like behavior, and topological regime separation—cannot be instantiated even in simple latent dynamical systems, the framework would lack methodological credibility.

We therefore use two toy experiments with distinct roles. The first tests *geometric identifiability*: whether user-conditioned latent dynamics generate re-identifiable trajectories above chance. The second tests *topological class separation*: whether convergent and cyclic regimes remain distinguishable after projection into a noisy higher-dimensional latent space.

4.1 Toy Experiment 1: Geometric Identifiability

The first toy experiment is a geometric sanity check for the notion of user-specific interaction structure. We consider a latent dynamical system of the form

$$x_{t+1} = (1 - \alpha - \beta)\, x_t + \alpha\, s_i + \beta\, d_m + \varepsilon_t, \tag{1}$$

where $x_t \in \mathbb{R}^d$ denotes the latent state at time t, s_i is a user-specific signature vector, d_m is a domain or model attractor, and $\varepsilon_t \sim \mathcal{N}(0, \sigma^2 I_d)$ is isotropic Gaussian noise. The interpretation is deliberately minimal: s_i models a user-conditioned directional bias in latent space, while d_m models task- or model-level regularization toward a shared region. This is a *controlled synthetic separability regime* and does not simulate a deployed user-model interaction.

4.1.1 Setup

We instantiate N=500 users and M=2 model domains in a d=512-dimensional embedding space. User signature vectors $s_i \sim \mathcal{N}(0, I_d)$ and domain attractors $d_m \sim \mathcal{N}(0, I_d)$ are sampled independently and ℓ_2-normalised. Trajectories are simulated with α=0.4, β=0.3, σ=0.15, and T=200 time steps. Empirical user signatures are estimated as

$$\hat{s}_i^{(m)} = \frac{1}{K} \sum_{t=T-K}^{T} x_t^{(i,m)} - \beta\, d_m, \quad K{=}40. \tag{2}$$

In the noise-free expectation, the deterministic fixed point of Eq. (1) is

$$x^* = \frac{\alpha\, s_i + \beta\, d_m}{\alpha + \beta}. \tag{3}$$

All experiments are repeated over ten independent random seeds {42, 123, 999, 2024, 7, 314, 1337, 77, 256, 88}.

4.1.2 Metrics

We evaluate three observables: (i) mean ℓ_2 distance between the empirical trajectory mean and the theoretical fixed point of Eq. (3); (ii) nearest-neighbor re-identification accuracy over N_{test}=500 fresh test queries per model; (iii) same-user versus different-user cosine similarity of empirical signatures across model domains.

4.1.3 Results

Table 1 reports summary statistics across all ten seeds.

Nearest-neighbor re-identification accuracy reaches 1.000 ± 0.000 across all seeds—a 500× improvement over the random baseline of 0.002 in this synthetic setting. Same-user cosine similarity (0.357 ± 0.002) is far above the near-zero cross-user similarity (0.002 ± 0.002), confirming geometric separability.

4.1.4 Robustness analysis

To preempt concerns about noise sensitivity, we vary $\sigma \in \{0.05, 0.15, 0.30, 0.50, 0.75, 1.00, 1.50, 2.00\}$ with all other parameters fixed (seed = 42).

Figure 2 reveals three regimes: a *stable regime* ($\sigma \leq 0.15$) with perfect re-identification; a *critical transition* at $\sigma \approx 0.20$–0.30 where accuracy drops sharply; and a *noise-dominated regime*

Table 1: Toy Experiment 1 — scaled results in the controlled synthetic separability regime (d=512, N=500, T=200, 10 seeds).

Metric	**Mean**	**Std**	**Min**	**Max**
Random baseline accuracy	0.0020	—	—	—
NN re-id accuracy	1.0000	0.0000	1.0000	1.0000
Mean attractor distance	0.7893	0.0009	0.7878	0.7907
Cosine (same user)	0.3570	0.0020	0.3538	0.3609
Cosine (different user)	0.0022	0.0019	-0.0011	0.0056

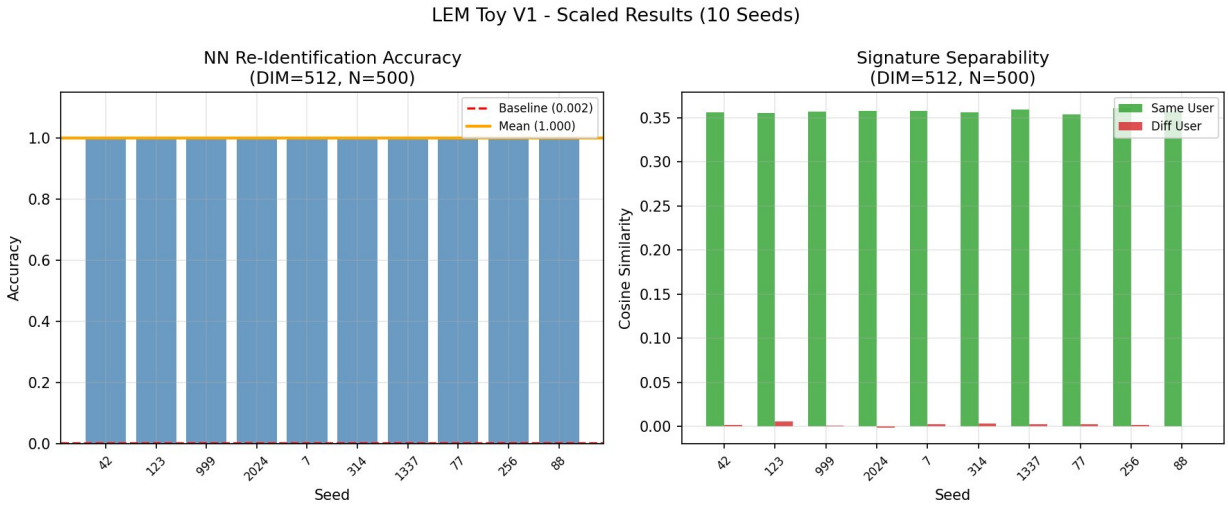

Figure 1: Toy Experiment 1. **Left:** NN re-identification accuracy across ten seeds (orange line = mean, red dashed = random baseline). **Right:** cosine similarity for same-user (green) versus different-user (red) signature pairs. All results are within the controlled synthetic separability regime described in Section 4.1.

($\sigma \geq 0.50$) where accuracy collapses to baseline. Attractor distance grows linearly with σ (slope ≈ 5.07), consistent with the theoretical prediction $\|\hat{x}^* - x^*\| \propto \sigma/(\alpha+\beta)$. Cross-user cosine similarity remains at ≈ 0 throughout, confirming that signature confusion rather than mere degradation drives the accuracy collapse.

The perfect re-identification at d=512 reflects the *blessing of dimensionality*: in high-dimensional spaces, random unit vectors are nearly orthogonal, rendering user signatures geometrically separable under moderate noise. This separability is a property of the high-dimensional synthetic construction and should not be interpreted as evidence that real deployed users are identifiable by such means. The sharp phase transition at $\sigma \approx 0.25$ defines a clear operating regime for the LEM and provides a principled criterion for when geometric identifiability can be expected to hold in controlled settings.

4.2 Toy Experiment 2: Topological Class Separation

The second toy experiment is the central numerical validation for the topological reformulation of LEM. Here, the aim is not to separate users geometrically, but to distinguish qualitatively different *dynamical regimes* through topological invariants.

Table 2: Robustness to noise level σ (d=512, N=500, seed = 42).

σ	NN Acc.	Attr. Dist.	Cos. (same)	Cos. (diff)
0.05	**1.0000**	0.3319	0.8037	0.0045
0.15	**1.0000**	0.7895	0.3545	0.0019
0.30	0.2300	1.5348	0.1219	0.0006
0.50	0.0200	2.5419	0.0469	0.0002
0.75	0.0050	3.8054	0.0207	0.0001
1.00	0.0033	5.0703	0.0112	0.0000
1.50	0.0033	7.6017	0.0043	0.0000
2.00	0.0017	10.1339	0.0018	0.0000

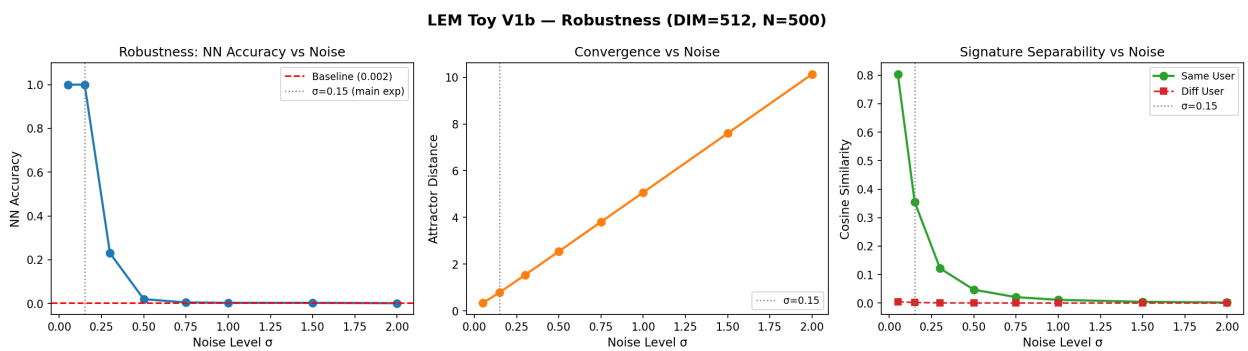

Figure 2: Robustness of LEM signatures to noise σ. **Left:** NN accuracy with phase transition at $\sigma \approx 0.25$. **Centre:** attractor distance (linear in σ). **Right:** cosine separability collapse.

4.2.1 Setup

We construct two synthetic trajectory classes:

1. a **convergent regime**, modeled as a stable sink in two dimensions:

$$\dot{x} = -\lambda x, \quad \dot{y} = -\lambda y, \quad \lambda = 0.5, \tag{4}$$

   initialised at $(x_0, y_0) = (2.0,\ 1.5)$;

2. a **cyclic regime**, modeled as a stable limit cycle via the Van der Pol oscillator:

$$\dot{x} = y, \quad \dot{y} = \mu(1 - x^2)y - x, \quad \mu = 1.5, \tag{5}$$

   initialised at $(x_0, y_0) = (0.1,\ 0.1)$.

Both systems are integrated over T=40 time units with Δt=0.1 (401 steps), then embedded into $\mathbb{R}^{16}$ via a random linear projection $A \in \mathbb{R}^{2\times 16}$ with additive Gaussian noise (σ=0.05). Persistent homology in degree H_1 is computed via the Vietoris–Rips filtration using `ripser` [6].

4.2.2 Metric

For each trajectory we compute the degree-1 persistence diagram $\mathrm{PD}(H_1)$. The key quantity is the *maximum persistence*

$$p_{\max} = \max_{(b,d)\,\in\,\mathrm{PD}(H_1)} (d - b),$$

where b and d denote the birth and death of a homological feature. A large $p_{\max}$ indicates a robust loop; values near zero indicate no persistent cyclic structure. We define the signal-to-noise ratio as

$$\mathrm{SNR} = \frac{p_{\max}^{\text{cyclic}}}{p_{\max}^{\text{convergent}} + \epsilon}, \quad \epsilon = 10^{-9}.$$

4.2.3 Results

Table 3: Toy Experiment 2 — topological class separation (d=16, T=40, seed = 1337).

Metric	**Convergent**	**Cyclic**
Max H_1 persistence	0.0326	**5.2649**
Mean H_1 persistence	0.0092	**0.1972**
SNR (cyclic / convergent)	**161.6**	

The cyclic regime produces a maximum H_1 persistence of 5.265, compared to 0.033 for the convergent regime—a ratio of 161.6×. A simple threshold classifier at $p_{\max} > 0.1$ separates the two regimes with zero error, requiring no learned parameters.

These results demonstrate that persistent homology reliably distinguishes convergent from cyclic LEM dynamics even after noisy high-dimensional embedding, without access to the original system parameters.

4.3 What the toy experiments do and do not establish

Taken together, the two toy experiments support a restricted interpretation of LEM.

Supported in the present paper.

- User-conditioned latent trajectories can be geometrically distinguishable above chance in a controlled synthetic separability regime.
- Convergent and cyclic trajectory classes can be topologically separated through persistent homology (SNR = 161.6).
- The vocabulary of attractors, signatures, and topological regime structure is operationalizable and falsifiable.

Not established in the present paper.

- That real frontier-model activation traces exhibit the same trajectory classes.
- That interaction signatures correspond to psychological or neuroscientific identity.
- That high-dimensional synthetic separability implies real-world user identifiability.
- That cross-version stability has already been demonstrated empirically.
- That LEM currently enables deployment-grade personalization, safety, or forensic attribution.

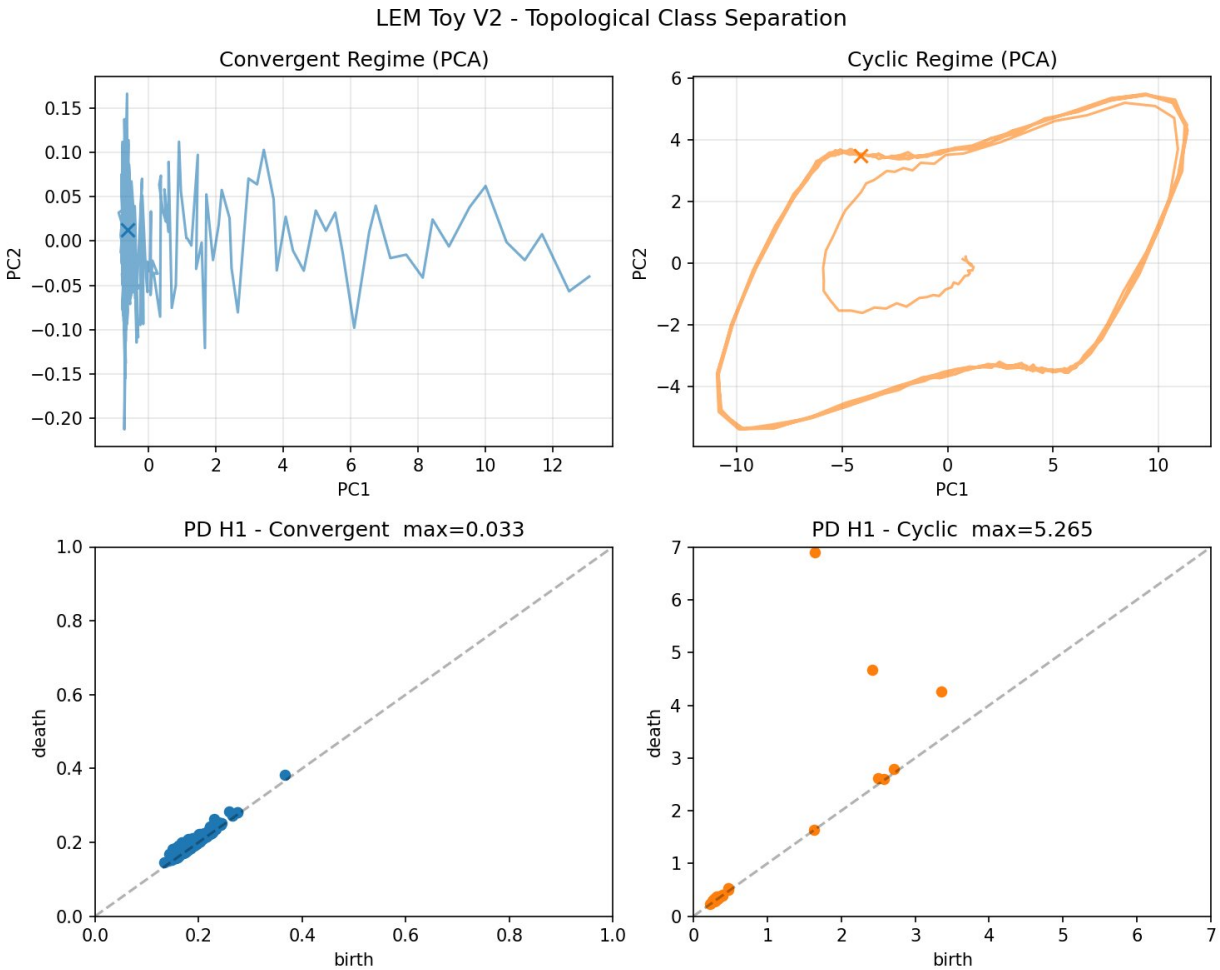

Figure 3: Toy Experiment 2: topological class separation. **Top:** PCA projections of the convergent (left, blue) and cyclic (right, orange) trajectories embedded in $\mathbb{R}^{16}$. **Bottom:** degree-1 persistence diagrams. The convergent diagram shows only diagonal noise ($p_{\max}$=0.033); the cyclic diagram contains a prominent off-diagonal generator ($p_{\max}$=5.265) corresponding to the limit-cycle topology. SNR = 161.6.

5 Related Work

LEM is related to several existing lines of research, but does not coincide exactly with any single one of them. Its closest neighbors lie in user modeling, latent behavioral representations, mechanistic interpretability, and topological analysis of neural representations.

5.1 User modeling and latent behavioral structure

A first neighboring tradition concerns latent user representations and personalization. In these approaches, user-specific state is typically modeled through compressed embeddings, memory summaries, or preference vectors. LEM is compatible with this general objective, but differs in one central respect: it treats repeated user–model interaction not primarily as a static user representation problem, but as a trajectory-level object. Related work on latent state persistence in LLMs has

raised important questions about whether models maintain stable internal representations across turns at all [7]. LEM does not presuppose that real LLMs do so; rather, it asks whether the structure of user-conditioned trajectories is detectable if such persistence were present.

A closer point of contact is recent work on persona vectors. Chen et al. identify directions in activation space that correspond to behavioral or character traits—such as sycophancy, hallucination propensity, or harmfulness—and demonstrate that these directions are causally effective: injecting or suppressing them during inference predictably alters model behavior [5]. This provides direct empirical evidence that the latent spaces through which LEM posits user-conditioned trajectories contain structured, behaviorally meaningful geometry. LEM builds on this foundation but shifts the focus from static trait directions to the temporal dynamics of repeated interaction trajectories and the possibility of deriving geometric and topological signatures from them.

5.2 Mechanistic interpretability and feature geometry

LEM is also related to mechanistic interpretability work that studies the internal organization of model representations. In particular, dictionary-learning and sparse-autoencoder work argues that model activations can be decomposed into more interpretable features than individual neurons alone [3, 10]. These results matter for LEM because they provide evidence that model representations are structured enough to support nontrivial latent analyses. A related direction is the literature on attribution graphs and circuit tracing, which models internal feature interactions as structured computational pathways [1]. LEM differs in being trajectory-centric rather than feature- or circuit-centric.

5.3 Topological analysis of representation structure

A third neighboring tradition concerns topological data analysis in machine learning and language models. The foundational framework for applying persistent homology to data analysis was established by Carlsson [4]. More recently, persistent homology has been used to study structure in learned representations, model layers, and reasoning traces [6, 9]. LEM draws directly from this methodological tradition, especially in its use of persistent H_1 structure to distinguish topologically trivial from recurrent interaction regimes.

The narrower novelty claim of LEM is therefore not that topology can be applied to neural representations in general, but that user–LLM interaction can be modeled as a dynamical trajectory whose regime class may be summarized through geometric and topological signatures.

5.4 Dynamical systems and attractor perspectives

A fourth neighboring tradition treats language models through the lens of dynamical systems. Ramsauer et al. show that Transformer attention corresponds to the update rule of modern Hopfield networks, whose energy landscape admits multiple attractor types [8]. Bai et al. demonstrate that hidden layers of deep sequence models can converge to fixed points, formalizing the notion of equilibrium computation in neural networks [2]. Most directly relevant to LEM, Wang et al. provide empirical evidence that iterative text generation in LLMs converges to attractor cycles, including two-period limit cycles under successive paraphrasing [11]. This establishes that attractor dynamics in language models are not merely a theoretical construct but an empirically observable phenomenon.

LEM differs from this line of work in three respects. First, it studies user-conditioned trajectories rather than generic model dynamics. Second, it operates in latent representation space rather than

in text space. Third, it employs topological analysis to distinguish regime types, whereas existing dynamical-systems work on LLMs has not used persistent homology for this purpose.

5.5 Positioning of the present contribution

The present paper should not be read as claiming to be the first connection between topology and language models, nor the first work on latent behavioral directions, nor the first dynamical-systems perspective on LLMs. Each of these components has established precedent. The narrower contribution of LEM is the synthesis: it introduces a single trajectory-based framework in which:

1. user-conditioned interaction is modeled dynamically (building on [11, 8, 2]);
2. attractor-like behavior becomes a first-class object of analysis, supported by evidence that latent spaces encode structured behavioral geometry [5, 3];
3. geometric identifiability and topological regime structure are treated jointly, using tools from persistent homology [4, 6].

No existing work in the surveyed literature combines all three elements. LEM is therefore best understood as a synthesis with a specific new emphasis: the study of user–LLM interaction through the combined lenses of trajectory dynamics, attractor structure, and topological signatures.

6 Discussion and Limitations

The main contribution of the present formulation is methodological. LEM is strongest when interpreted not as a completed explanatory theory, but as a compact research program that introduces a trajectory-based object of analysis and proposes geometric and topological tools for studying it.

The toy models support two limited claims:

1. user-conditioned trajectories can be geometrically identifiable above chance in a controlled synthetic separability regime;
2. convergent and cyclic trajectory classes can be topologically separated through persistent homology.

These findings do not establish that real frontier-model interaction traces behave in the same way. They instead show that the theoretical vocabulary of LEM is operationalizable and falsifiable in controlled synthetic settings.

It bears emphasis that the perfect re-identification observed in Toy Experiment 1 is a consequence of the high-dimensional synthetic construction, not evidence for real-world user fingerprinting. In d=512 dimensions, independently sampled random unit vectors are near-orthogonal by concentration-of-measure arguments, making separation trivially achievable in the absence of real-world confounds such as overlapping behavioral styles, model-induced compression, or adversarial noise. The value of Toy Experiment 1 lies not in the magnitude of the accuracy, but in confirming that the attractor-based update rule produces stable, structured trajectories in the controlled regime.

This work has several further limitations. First, all numerical validation is based on synthetic toy dynamics rather than internal activation traces from real language models. Second, the notion of an interaction signature is operational and representational, not psychological or neuroscientific. Third, the current paper does not establish cross-version or cross-model stability. Fourth, no claim

is made here that topological signatures are already useful for deployed personalization, safety, or forensic attribution.

The paper deliberately avoids stronger claims about cognition, consciousness, or human identity. Its scope is restricted to a formal and computational framework for analyzing interaction dynamics.

7 Future Work

There are four immediate next steps:

1. extend the toy-model logic to real multi-turn activation traces from open-weight language models;
2. compare geometric and topological signatures more systematically, including bottleneck and Wasserstein diagram distances;
3. operationalize cross-version stability through diagram-based distances or related invariants;
4. connect LEM more explicitly to current interpretability workflows, including feature-based and persona-based latent analyses.

More ambitious directions, including cross-model user continuity, privacy implications, and robust trajectory-level defenses, should be treated as future hypotheses rather than present conclusions.

The framework documentation and code are available at `https://github.com/LAKITALKS/LEM`. A citable record is archived at Zenodo (DOI: 10.5281/zenodo.17969735).

8 Conclusion

LEM provides a compact and testable framework for analyzing user–LLM interaction dynamics as trajectories rather than isolated states. Its central contribution is to combine attractor-based intuition with a topological reformulation that makes convergent and cyclic regimes measurable in principle. The current paper presents a first version of this program: conceptual formalization plus toy-model numerical validation. Whether these ideas scale to real language-model internals remains an open empirical question. However, even at the present stage, LEM provides a clearer basis for asking that question in a controlled and scientifically tractable way.

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